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High-voltage nano-sheet transistor

Abstract

The present disclosure is directed to methods for the formation of high-voltage nano-sheet transistors and low-voltage gate-all-around transistors on a common substrate. The method includes forming a fin structure with first and second nano-sheet layers on the substrate. The method also includes forming a gate structure having a first dielectric and a first gate electrode on the fin structure and removing portions of the fin structure not covered by the gate structure. The method further includes partially etching exposed surfaces of the first nano-sheet layers to form recessed portions of the first nano-sheet layers in the fin structure and forming a spacer structure on the recessed portions. In addition, the method includes replacing the first gate electrode with a second dielectric and a second gate electrode, and forming an epitaxial structure abutting the fin structure.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is a divisional of U.S. patent application Ser. No. 16/916,951, titled “High-Voltage Nano-sheet Transistor,” filed on Jun. 30, 2020, which is incorporated herein by reference in its entirety.

BACKGROUND
(1) One time programmable (OTP) memory is a type of non-volatile memory (NVM) that permits data to be written to memory only once. Once the memory has been programmed, it retains its

value upon loss of power. OTP memory is used in applications where reliable and repeatable reading of data is required. Examples include boot code, encryption keys and configuration parameters for analog, sensor or display circuitry. OTP NVM offers a low power, small area footprint memory structure. OTP memory is applicable in products from microprocessors and display drivers to Power Management Integrated Circuits (PMICs).

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures.

(2) FIG. **1** is a cross-sectional view of a high-voltage nano-sheet field effect transistor on a substrate, in accordance with some embodiments.

(3) FIG. **2** is a cross-sectional view of a gate-all-around nano-sheet field effect transistor on a substrate, in accordance with some embodiments.

(4) FIGS. **3**A and **3**B are flow diagrams of a method for the concurrent fabrication of a high-voltage nano-sheet field effect transistor and a gate-all-around nano-sheet field effect transistor on a common substrate, in accordance with some embodiments.

(5) FIGS. **4-10** are isometric views of intermediate structures during the concurrent fabrication of a high-voltage nano-sheet field effect transistor and a gate-all-around nano-sheet field effect transistor on a common substrate, in accordance with some embodiments.

(6) FIGS. **11-18** are cross-sectional views of intermediate structures during the concurrent fabrication of a high-voltage nano-sheet field effect transistor and a gate-all-around nano-sheet field effect transistor on a common substrate, in accordance with some embodiments.

(7) FIG. **19** is a cross-sectional view of a high-voltage nano-sheet field effect transistor and a gate-all-around nano-sheet field effect transistor formed on a common substrate, in accordance with some embodiments

(8) FIGS. **20**A and **20**B are circuit layouts of voltage transistors and bit-cell arrays, in accordance with some embodiments.

DETAILED DESCRIPTION

(9) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed that are between the first and second features, such that the first and second features are not in direct contact.

(10) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(11) The term “nominal” as used herein refers to a desired, or target, value of a characteristic or parameter for a component or a process operation, set during the design phase of a product or a process, together with a range of values above and/or below the desired value. The range of values is typically due to slight variations in manufacturing processes or tolerances.

(12) In some embodiments, the terms "about" and "substantially" can indicate a value of a given quantity that varies within 5% of the value (e.g., ±1%, ±2%, ±3%, ±4%, ±5% of the value). These values are merely examples and are not intended to be limiting. It is to be understood that the terms "about" and "substantially" can refer to a percentage of the values as interpreted by those skilled in relevant art(s) in light of the teachings herein.
(13) The term "vertical," as used herein, means nominally perpendicular to the surface of a substrate.
(14) The term "insulating layer", as used herein, refers to a layer that functions as an electrical insulator (e.g., a dielectric layer).
(15) Gate-all-around (GAA) field effect transistors (GAA-FETs), such as nano-sheet or nano-wire GAA-FETs, have an improved gate control over their channel region compared to other types of FETs whose gate structure covers sidewall portions and top surfaces of their semiconductor fin structure. Due to their gate-all-around geometry, GAA nano-sheet or nano-wire FETs achieve larger effective channel widths and higher drive currents. With memory manufacturers shifting towards GAA-FETs, there is an increasing interest for nano-sheet or nano-wire FET structures capable of operating at high voltages (e.g., about 5 Volts), such as in one time programmable (OTP) memory cells. OTP memory is a type of non-volatile memory (NVM) that permits data to be written to memory only once. Once the memory has been programmed, it retains its value upon loss of power. OTP memory is used in applications where reliable and repeatable reading of data is required. Examples include boot code encryption keys and configuration parameters for analog, sensor or display circuitry used in a range of electronic devices including, but not limited to, smart phones, gaming consoles, tablets, wearable electronic devices, and other personal devices.
(16) The embodiments described herein are directed to methods for the simultaneous fabrication of nano-sheet or nano-wire FETs for high-voltage applications, such as in OTP memory cells, with low-voltage GAA FETs. For example, high-voltage nano-sheet or nano-wire FETs, collectively referred to herein as HV-NS FETs, can be formed alongside low-voltage GAA FETs (LV-GAA FETs) which operate at about 1.8 Volts or less. The HV-NS FETs as described herein include a gate dielectric stack with a silicon oxide layer at a thickness between about 1.5 nm and about 3.5 nm and a high-k dielectric layer at a thickness of about 1.5 nm disposed thereon. In some embodiments, a silicon oxide layer that is thinner than about 1.5 nm cannot withstand an operating voltage greater than about 5 Volts. A silicon oxide layer that is thicker than about 3.5 nm requires a higher operating voltage than about 5 Volts, which increases the power consumption of the HV-NS FET. In some embodiments, the HV-NS FETs include the same metal gate electrodes found in the LV-GAA FETs. According to some embodiments, the HV-NS FETs, contrary to LV-GAA FETs, include a fin structure with alternating silicon (Si) and silicon-germanium (SiGe) nano-sheet layers. In some embodiments, and during operation, current conduction in the HV-NS FETs occurs within the silicon nano-sheet layers. Further, both HV-NS FETs and LV-GAA FETs share similar S/D epitaxial structures.
(17) According to some embodiments, FIG. **1** is a cross-sectional view of a HV-NS FET **100** formed on a substrate **110**. According to some embodiments, HV-NS FET **100** is capable of operating at a voltage of about 5 Volts. In some embodiments, FIG. **1** shows a partial cross-sectional view of substrate **110** and other portions of substrate **110** may not be shown. For example, other portions of substrate **110** can include additional HV-NS FETs and LV-GAA FETs, which are not shown. These additional HV-NS FETs and LV-GAA FETs may be formed next to HV-NS FET **100** or on other areas of substrate **110** away from HV-NS FET **100**.
(18) According to some embodiments, HV-NS FET **100** includes a fin structure formed by alternating SiGe nano-sheet layers or nano-wires **120** and Si nano-sheet layers or nano-wires **130**. Source/drain (S/D) epitaxial layers **140** of HV-NS FET **100** are in physical contact with Si nano-sheet layers or nano-wires **130** and isolated from SiGe nano-sheet layers or nano-wires **130** through spacer structures **150**. Gate structure **160** of HV-NS FET **100** surrounds the fin structure of SiGe

nano-sheet layers or nano-wires **120** and Si nano-sheet layers or nano-wires **130** and includes a silicon oxide (SiO.sub.2) dielectric layer **160***a* ("gate dielectric **160***a*") with a thickness between about 1.5 nm and about 3.5 nm, a high-k dielectric **160***b* (e.g., a hafnium-based dielectric) ("high-k gate dielectric **160***b*"), and a gate electrode **160***c*. In some embodiments, gate electrode **160***c* further includes protective layers for high-k gate dielectric **160***b*, work function layers (e.g., metallic layers) and metal fill layers not shown in FIG. **1**. In some embodiments, gate spacers **170** are formed on sidewall surfaces of high-k gate dielectric **160***b* and on end-portions of gate dielectric **160***a* as shown in FIG. **1**. In some embodiments, gate spacers **170** include silicon nitride (Si.sub.3N.sub.4 or "SiN"), silicon carbon nitride (SiCN), or silicon carbon oxy-nitride (SiCON) material. In some embodiments, gate spacers **170** facilitate the formation of gate structure **160** and the fin structure formed by the SiGe and Si nano-sheet layers. HV-NS FET **100** is isolated from neighboring HV-NS FETs or LV GAA FETs (not shown) through interlayer dielectric (ILD) **180**, which includes one or more layers of dielectric material. In some embodiments, ILD **180** is a silicon oxide based dielectric, which further includes nitrogen, hydrogen, carbon, or combinations thereof.

(19) In some embodiments, substrate **110** is a top semiconductor layer of a bulk wafer or a top semiconductor layer of a silicon on insulator (SOI) wafer. In some embodiments, substrate **110** includes crystalline Si or another elementary semiconductor, such as germanium (Ge). Alternatively, substrate **110** may include (i) a compound semiconductor like silicon carbide (SiC), gallium arsenide (GaAs), gallium phosphide (GaP), indium phosphide (InP), indium arsenide (InAs), and/or indium antimonide (InSb); (ii) an alloy semiconductor like silicon germanium (SiGe), gallium arsenide phosphide (GaAsP), aluminum indium arsenide (AlInAs), aluminum gallium arsenide (AlGaAs), gallium indium arsenide (GaInAs), gallium indium phosphide (GaInP), and/or gallium indium arsenide phosphide (GaInAsP); or (iv) combinations thereof.

(20) For example purposes, substrate **110** will be described in the context of crystalline Si bulk wafer. Based on the disclosure herein, other substrates or other substrate materials, as discussed above, can be used. These other substrates or substrate materials are within the spirit and scope of this disclosure.

(21) In some embodiments, for a p-type HV-NS FET, S/D epitaxial layers **140** include boron-doped (B-doped) SiGe, B-doped germanium (Ge), B-doped germanium-tin (GeSn), or combinations thereof. In some embodiments, for an n-type HV-NS FET, S/D epitaxial layers **140** include arsenic (As) doped or phosphorous (P) doped Si, carbon doped silicon (Si:C), or combinations thereof. In some embodiments, S/D epitaxial layers **140** include two or more epitaxially grown layers, which are not shown in FIG. **1** for simplicity. In some embodiments, S/D epitaxial layers **140** are grown from exposed sidewall surfaces of semiconductor nano-sheets layers or nano-wires **130** and surfaces of substrate **110**—for example, semiconductor nano-sheets or nano-wires **130** and substrate **110** function as seed layers for the S/D epitaxial layers. In some embodiments, S/D epitaxial layers **140** have a diamond shape, a hexagonal shape, or any other faceted shape.

(22) In some embodiments, SiGe/Si nano-sheets or nano-wires **120** and **130** are referred to as "nano-sheets" when their width (e.g., along the y-direction) is substantially different from their height (e.g., along the z-direction)—for example, when their width is larger/narrower than their height. In some embodiments, SiGe and Si nano-sheets or nano-wires **120** and **130** are referred to as "nano-wires" when their width is substantially equal to their height. By way of example and not limitation, SiGe and Si nano-sheets or nano-wires **120** and **130** will be described in the context of nano-sheets. Based on the disclosure herein, SiGe and Si nano-wires, as discussed above, are within the spirit and the scope of this disclosure.

(23) In some embodiments, each nano-sheet layer has a vertical thickness or height (e.g., along the z-direction) between about 5 nm and about 8 nm, and a width along the y-direction between about 8 nm and about 25 nm. In some embodiments, HV-NS FET **100** includes between 2 and 8 individual nano-sheet layers depending on the FET's desired electrical characteristics. This is not

limiting and additional nano-sheet layers are possible. In some embodiments, Si nano-sheet layers **130** are lightly doped or undoped. If lightly doped, the doping level of Si nano-sheet layers **130** is less than about 10.sup.13 atoms/cm.sup.3. In some embodiments, SiGe nano-sheet layers **120** include a Ge atomic concentration between about 20% and about 30%.

(24) As discussed above, edge portions of SiGe nano-sheet layers **120** are covered by spacer structures **150**. In some embodiments, spacer structures **150**, like gate spacers **170**, include a nitride, such as SiN, SiCN, or SiCON. In some embodiments, the width of spacer structures **150** along the x-direction ranges between about 5 nm and about 10 nm. As shown in FIG. **1**, spacer structures **150** are interposed between SiGe nano-sheet layers **120** and S/D epitaxial layers **140** to electrically isolate them.

(25) According to some embodiments, FIG. **2** is a cross-sectional view of a LV-GAA FET **200** formed on substrate **110**. According to some embodiments, LV-GAA FET **200** is configured to operate at a lower voltage—for example at a voltage of about 1.8 Volts or less. As shown in FIG. **2**, LV-GAA FET **200** shares similar components with HV-NS FET **100** shown in FIG. **1**. However, notable differences include the absence of SIG nano-sheet layers **120** from the fin structure of LV-GAA FET **200** and the absence of gate dielectric **160***a* below high-k gate dielectric **160***b* in gate structure **160**. More specifically, SiGe nano-sheet layers **120** have been replaced with gate structure **160**, which includes an additional interfacial layer (IL) **160***d*. As a result, the space between semiconductor nano-sheet layers **130** is occupied by the layers of gate structure **160**—for example, IL **160***d*, high-k gate dielectric **160***b*, and gate electrode **160***c*. In some embodiments, gate structure **160** covers a middle section of Si nano-sheet layers **130**. Therefore, gate structure **160** surrounds Si nano-sheet layers **130** and forms a GAA structure. As shown in FIG. **2**, the layers of gate structure **160** disposed between Si nano-sheet layers **130** are isolated from S/D epitaxial layers **140** via spacer structures **150**.

(26) Due to the above structural differences, LV-GAA FET **200** and HV-NS FET **100** operate differently. For example, LV-GAA FET **200** is a GAA device while HV-NS FET **100** functions as a pseudo finFET device since gate structure **160** covers primarily sidewall surfaces of Si nano-sheet layers **130** and a top surface of the uppermost Si nano-sheet layer **130**. Therefore, HV-NS FET **100** does not feature a GAA configuration, nor does it operate as a GAA device. This is intentional because HV-NS FET **100**, unlike LV-GAA FET **200**, does not require improved gate control over the channel region. Further the configuration of HV-NS FET **100** as described above reduces the manufacturing cost. In some embodiments, HV-NS FET **100** and LV-GAA FET **200** are formed concurrently on substrate **110**.

(27) According to some embodiments, FIGS. **3**A and **3**B show flow charts of a fabrication method **300** describing the formation of HV-NS FET **100** shown in FIG. **1** and LV GAA FET **200** shown in FIG. **2** on a common substrate, such as substrate **110**. Method **300** is independent of the substrate used—for example, method **300** can be used to form HV-NS FETs and LV GAA FET **200** on SOI substrates or other types of substrates. Other fabrication operations can be performed between the various operations of method **300** and are omitted merely for clarity. Additionally, some of the operations may be performed simultaneously, or in a different order than the ones shown in FIGS. **3**A and **3**B. In some embodiments, one or more other operations may be performed in addition to or in place of the presently described operations. For illustrative purposes, method **300** is described with reference to the embodiments shown in FIGS. **1**, **2**, and **4-17**.

(28) Method **300** begins with operation **305** and the process of forming alternating SiGe and Si nano-sheet layers on a substrate (e.g., substrate **110**). The formation of these layers is common for both HV-NS FET **100** shown in FIG. **1** and LV GAA FET **200** shown in FIG. **2**. FIG. **4** is a partial isometric view of SiGe nano-sheet layers **120** and Si nano-sheet layers **130** deposited on substrate **110** according to operation **305**. In some embodiments, SiGe nano-sheet layers **120** and Si nano-sheet layers **130** can be grown directly on substrate **110** (e.g., an SOI substrate or a bulk substrate) with a chemical vapor deposition (CVD) process using silane (SiH.sub.4), disilane

(Si.sub.2H.sub.6), germane (GeH.sub.4), digermane (Ge.sub.2H.sub.6), dichlorosilane (SiH.sub.2Cl.sub.2), other suitable gases, or combinations thereof. As discussed above, SiGe nano-sheet layers **120** contain between about 20% and about 30% Ge while Si nano-sheet layers **130** are substantially germanium-free. In some embodiments, SiGe nano-sheet layers **120** and/or Si nano-sheet layers **130** can be doped.

(29) In some embodiments, the thickness of SiGe nano-sheet layers **120** defines the spacing between every other Si nano-sheet layer **130**, and similarly the thickness of Si nano-sheet layers **130** defines the spacing between every other SiGe nano-sheet layer **120** in the stack. For example, in referring to FIG. **5**, which is a magnified view of section **400** shown in FIG. **4**, thickness **120**(T) of SiGe nano-sheet layer **120** can be used to define the spacing of Si nano-sheet layers **130**. As discussed above, the thickness of each nano-sheet layer can range from about 5 nm to about 8 nm. Since the SiGe and Si nano-sheet layers are grown individually, the SiGe nano-sheet layers **120** and the Si nano-sheet layers **130** can have a similar or different thickness from one another. Further, each of the SiGe nano-sheet layers can have a similar or different thickness from one another, and similarly each of the Si nano-sheet layers can have a similar or different thickness from one another. The aforementioned thickness permutations are within the spirit and the scope of this disclosure.

(30) In referring to FIG. **3**A, method **300** continues with operation **310** and the process of patterning the SiGe and Si nano-sheet layers to form a vertical nano-sheet layer structure. The vertical nano-sheet structure according to operation **310** is formed concurrently for HV-NS FET **100** shown in FIG. **1** and LV GAA FET **200** shown in FIG. **2**. In some embodiments, the vertical nano-sheet layer structure can be formed as follows. In referring to FIGS. **6** and **7**A, a photoresist layer is spin-coated over the uppermost Si nano-sheet layer **130** and subsequently patterned to form patterned photoresist structure **600**. Patterned photoresist structure **600** functions as an etch mask in a subsequent etching process during which portions of Si nano-sheet layers **130** and SiGe nano-sheet layers **120** not covered by patterned photoresist structure **600** (e.g., not masked) are removed to form a vertical nano-sheet layer structure **700** shown in FIG. **7**A. After the formation of vertical nano-sheet layer structure **700**, patterned photoresist structure **600** is removed from vertical nano-sheet layer structure **700** with a wet etching process. In some embodiments, during the aforementioned patterning process, substrate **110** is also patterned to form pedestal structure **110***p*. According to some embodiments, pedestal structure **110***p* facilitates the formation of a shallow trench isolation (STI) structure **710** shown in FIG. **7**B. In some embodiments, during operation **310**, multiple vertical nano-sheet layer structures (e.g., like vertical nano-sheet layer structure **700**) are formed for HV-NS FETs and LV GAA FETs on substrate **110**.

(31) In some embodiments, width **600***w* of patterned photoresist structure **600** defines the width of vertical nano-sheet layer structure **700** along the y-dimension, which subsequently defines the width of the channel region in HV-NS FET **100** shown in FIG. **1** and LV GAA FET **200** shown in FIG. **2**. Further, by controlling width **600***w* of pattern photoresist structure **600**, vertical nano-sheet layer structure **700** with different widths can be formed on substrate **110**. For example, if desired, HV-NS FET **100** and LV GAA FET **200** can be formed with different nano-sheet layer widths at any desired location on substrate **110**. In some embodiments, width **600***w* for HV-NS FET **100** shown in FIG. **1** ranges between about 8 nm and about 25 nm. In some embodiments, a width **600***w* less than about 8 nm reduces the drive current during the FET operation, while a width **600***w* greater than about 25 nm reduces the gate control over the channel region—both of which are not desirable.

(32) After removing patterned photoresist structure **600** from vertical nano-sheet layer structure **700**, STI structure **710** is formed on a top surface of substrate **110**. In some embodiments, to form STI structure **710**, STI material (e.g., a silicon oxide based dielectric) is blanket deposited over vertical nano-sheet layer structure **700** and substrate **110**. The as-deposited STI material can be subsequently planarized (e.g., with a chemical mechanical polishing (CMP) process) so that the top

surface of the STI material is coplanar with the top surface of vertical nano-sheet layer structure **700**. The planarized STI material is subsequently etched-back so that the resulting STI structure **710** has a height substantially equal to pedestal structure **110***p*, as shown in FIG. **7**B. In some embodiments, vertical nano-sheet layer structure **700** protrudes from STI structure **710** so that STI structure **710** does not cover any sidewall portion of vertical nano-sheet layer structure **700** as shown in FIG. **7**B.

(33) In referring to FIGS. **3**A and **8**, method **300** continues with operation **315** and the process of forming a sacrificial gate structure **800** on vertical nano-sheet layer structure **700**. At this fabrication stage, operation **315** is common between HV-NS FET **100** and LV GAA FET **200** shown respectively in FIGS. **1** and **2**. In some embodiments, sacrificial gate structure **800** includes a sacrificial polysilicon gate electrode **805** and gate dielectric **160***a*. In some embodiments, polysilicon gate electrode **805** and gate dielectric **160***a* can be blanket deposited to cover side and top surface portions of vertical nano-sheet layer structure **700** and the top surface of STI structure **710**. The blanket deposited polysilicon gate electrode **805** and gate dielectric **160***a* are subsequently patterned to form sacrificial gate structure **800** shown in FIG. **8**. In some embodiments, gate dielectric **160***a* includes silicon oxide (SiO.sub.2) or silicon-oxynitride silicon oxide (SiON) interposed between polysilicon gate electrode **805** and vertical nano-sheet layer structure **700**. For example, gate dielectric **160***a* is formed prior to polysilicon gate electrode **805**. In some embodiments, prior to the formation of gate dielectric **160***a*, an interfacial layer (not shown) is formed on fin structure **700** by ozone (O.sub.3) exposure at a thickness of about 1 nm. The formation of the interfacial layer can also include a pre-clean process that removes unwanted native oxides from the surfaces of fin structure **700** prior to the ozone exposure. By way of example and not limitation, the pre-clean process can include an SC-1 clean (e.g., a mixture of water, ammonia, and hydrogen peroxide) and an SC-2 clean (e.g., a mixture of water, hydrochloric acid, and hydrogen peroxide). In some embodiments, gate dielectric **160***a* is grown in a furnace or deposited by plasma-enhanced atomic layer deposition (PEALD) at a thickness between about 1.5 nm and about 3.5 nm. In some embodiments, a gate dielectric that is thinner than about 1.5 nm cannot withstand operating voltages of about 5 Volts. Respectively, a gate dielectric that is thicker than about 3.5 nm requires a higher operating voltage than about 5 Volts, which increases the power consumption of HV-NS FET **100**.

(34) As discussed above with respect to in FIGS. **1** and **2**, gate dielectric **160***a* remains as part of the gate structure **160** in HV-NS FET **100** and is replaced by IL **160***d* in LV GAA FET **200**.

(35) In some embodiments, sacrificial gate structure **800** is formed perpendicular to a length (e.g., the longest dimension) of vertical nano-sheet layer structure **700**—for example, along the y-dimension and perpendicular to the x-direction). Further, sacrificial gate structure **800** does not cover the entire length of vertical nano-sheet layer structure **700**. In some embodiments, as shown in FIG. **8**, edge portions of vertical nano-sheet layer structure **700** are not covered (e.g., not masked) by sacrificial gate structure **800**. For example, width **700**L is greater than width **800**L (**700**L>**800**L). By way of example and not limitation, **800**L is referred to as physical gate length and ranges between about 50 nm and about 150 nm.

(36) Sidewalls of sacrificial gate structure **800** are covered by gate spacers **170**, which are also shown in FIGS. **1** and **2**. In some embodiments, gate spacers **170** are not removed by a replacement gate process during which a portion of sacrificial gate structure **800** is replaced by gate structure **160**. In some embodiments, a top surface of polysilicon gate electrode **805** is covered with a gate capping or protective layer **815**. In some embodiments, gate capping layer **815** can be a stack of layers. For example, gate capping layer **815** may include an oxide layer (e.g., silicon oxide) and a nitride layer (e.g., SiN, SiON, SiOCN, etc.) not separately shown in FIG. **8**. In some embodiments, gate capping layer **815** and gate spacers **170** protect sacrificial gate structure **800** from subsequent processing operations.

(37) In referring to FIG. **3**A, method **300** continues with operation **320** and the process of removing

(e.g., trimming) portions vertical nano-sheet layer structure **700** not covered by sacrificial gate structure **800** as shown in FIG. **9**. In some embodiments, operation **320** is common between HV-NS FET **100** and LV GAA FET **200** shown respectively in FIGS. **1** and **2**. In some embodiments, the removal process involves a dry etching process, a wet etching process, or a combination thereof selective towards Si nano-sheet layers **130** and SiGe nano-sheet layers **120**. In some embodiments, the dry etching process includes etchants with an (i) oxygen-containing gas; (ii) methane (CH.sub.4); (iii) a fluorine-containing gas (e.g., carbon tetrafluoride (CF.sub.4), sulfur hexafluoride (SF.sub.6), difluoromethane (CH.sub.2F.sub.2), trifluoromethane (CHF.sub.3), and/or hexafluoroethane (C.sub.2F.sub.6)); (iv) a chlorine-containing gas (e.g., chlorine (Cl.sub.2), chloroform (CHCl.sub.3), carbon tetrachloride (CCl), and/or boron trichloride (BCl.sub.3)); (v) a bromine-containing gas (e.g., hydrogen bromide (HBr) and/or bromoform (CHBr.sub.3)); (vi) an iodine-containing gas; (vii) other suitable etching gases and/or plasmas; or combinations thereof. The wet etching process can include etching in diluted hydrofluoric acid (DHF), potassium hydroxide (KOH) solution, ammonia, a solution containing hydrofluoric acid (HF), nitric acid (HNO.sub.3), acetic acid (CH.sub.3COOH), or combinations thereof. In some embodiments, the etching chemistry does not substantially etch STI structure **710**, gate capping layer **815**, and gate spacers **170**. In some embodiments, STI structure **710** is used as an etch stop layer for the etching process described above. As shown in FIG. **9**, the removal process of operation **320** exposes pedestal structure **110*p*** of substrate **110**.

(38) In referring to FIG. **3**A, method **300** continues with operation **325** and the process of partially etching the SiGe nano-sheet layers **120** from vertical nano-sheet layer structure **700**. In some embodiments, operation **325** is common between HV-NS FET **100** and LV GAA FET **200** shown respectively in FIGS. **1** and **2**. According to some embodiments, FIG. **10** shows the structure of FIG. **9** after operation **325** where exposed edges of SiGe nano-sheet layers **120** have been laterally recessed along the x-direction. In some embodiments, FIG. **11** is a cross-sectional view of the structure shown in FIG. **10** along plane **1000**.

(39) According to some embodiments, exposed edges of SiGe nano-sheet layers **120** are recessed (e.g., partially etched) by an amount A that ranges between about 5 nm and about 10 nm along the x-direction as shown in FIG. **11**. In some embodiments, the recess in SiGe nano-sheet layers **120** can be achieved with a dry etching process that is selective towards SiGe. For example, halogen-based chemistries exhibit high etch selectivity towards Ge and low towards Si. Therefore, halogen gases etch Ge-containing layers (e.g., SiGe nano-sheet layers **120**) at a higher etching rate than substantially Ge-free layers (e.g., Si nano-sheet layers **130**). In some embodiments, the halogen-based chemistries include fluorine-based and/or chlorine-based gasses. Alternatively, a wet etching chemistry with high selectivity towards SiGe can be used. In some embodiments, a wet etching chemistry may include a mixture of sulfuric acid (H.sub.2SO.sub.4) and hydrogen peroxide (H.sub.2O.sub.2) (SPM) and diluted hydrofluoric acid (DHF), or a mixture of ammonia hydroxide with H.sub.2O.sub.2 and water (APM). The aforementioned etching processes are timed so that the desired amount of SiGe is removed.

(40) In some embodiments, SiGe nano-sheet layers **120** with a higher Ge atomic concentration have a higher etching rate than SiGe nano-sheet layers **120** with a lower Ge atomic concentration. Therefore, the etching rate of the aforementioned etching processes can be adjusted by modulating the Ge atomic concentration (e.g., the Ge content) in SiGe nano-sheet layers **120**. As discussed above, the Ge content in SiGe nano-sheet layers **120** can range between about 20% and about 30%. Consequently, a SiGe nano-sheet layer having about 20% Ge is etched slower than a SiGe nano-sheet layer having about 30% Ge. Therefore, the Ge concentration can be adjusted accordingly to achieve the desired etching rate and selectivity between SiGe nano-sheet layers **120** and Si nano-sheet layers **130**.

(41) In some embodiments, a Ge concentration below about 20% does not provide adequate selectivity between SiGe nano-sheet layers **120** and Si nano-sheet layers **130**. For example, the

etching rate between SiGe nano-sheet layers **120** and Si nano-sheet layers **130** becomes substantially similar to one another and both types of nano-sheet layers are etched during the etching process. On the other hand, for Ge concentrations higher than about 30%, Ge atoms can out-diffuse from SiGe nano-sheet layers **120** towards Si nano-sheet layers **130** (e.g., during growth) and change the selectivity between SiGe nano-sheet layers **120** and Si nano-sheet layers **130** during etching. Since Ge out-diffusion cannot be controlled, Ge concentrations higher than about 30% can result in unpredictable etching amounts.

(42) In referring to FIG. **3**A, method **300** continues with operation **330** and the process of depositing a capping layer on vertical nano-sheet layer structure **700**. In some embodiments, operation **330** is common between HV-NS FET **100** and LV GAA FET **200** shown respectively in FIGS. **1** and **2**. In some embodiments, the capping layer is blanket deposited over the entire structure shown in FIG. **11**. For example, in referring to FIG. **12**, capping layer **1200** of operation **330** is deposited on the exposed surfaces of substrate **110**, vertical nano-sheet layer structure **700**, gate spacers **170**, and gate capping layer **815**. In some embodiments, capping layer **1200** is deposited at a thickness between about 5 nm and about 10 nm or any other thickness to substantially fill recess amount A shown in FIG. **11**. In some embodiments, capping layer **1200** includes a silicon-based dielectric, such as SiN, SiOCN, SiCN, or SiON. In some embodiments, capping layer **1200** can be deposited with a plasma-enhance atomic layer deposition (PEALD) process or another suitable method capable of depositing conformal layers. As shown in FIG. **12**, capping layer **1200** fills the space formed by the recessed edge portions of SiGe nano-sheet layers **120**. Because of capping layer **1200** deposition, sidewall surfaces of vertical nano-sheet layer structure **700** are no longer exposed.

(43) In referring to FIG. **3**B, method **300** continues with operation **335** and the process of etching portions of capping layer **1200** to form spacer structures **150** on etched portions of SiGe nano-sheet layers **120**. In some embodiments, operation **335** is common between HV-NS FET **100** and LV GAA FET **200** shown respectively in FIGS. **1** and **2**. In some embodiments, capping layer **1200** can be etched with a dry etching process or a wet etching process. By way of example and not limitation, a dry etching process includes a combination of a organofluorine chemistry, such as sulfur hexafluoride (SF_6), carbon tetrafluoride (CF_4), nitrogen trifluoride (NF_3), fluoroform (CHF_3), 1,1-difluoroethane (CH_3CHF_2), or combination thereof. On the other hand, a wet etching chemistry can include, for example, hot phosphoric acid (H_3PO_4).

(44) As discussed above, operation **335** results in the formation of spacer structures **150**. According to some embodiments, spacer structures **150** electrically isolate SiGe nano-sheet layers **120** from S/D epitaxial layers **140** in HV-NS FET **100** (e.g., shown in FIG. **1**) and from gate structure **160** in LV GAA FET **200** (e.g., shown in FIG. **2**).

(45) Referring to FIGS. **3**B and **13**, method **300** continues with operation **340** and the process of forming S/D epitaxial layers **140** on exposed surfaces of vertical nano-sheet layer structure **700**. In some embodiments, operation **340** is common between HV-NS FET **100** and LV GAA FET **200** shown respectively in FIGS. **1** and **2**. In some embodiments, S/D epitaxial layers **140** are grown with a CVD process similar to the one used to form SiGe nano-sheet layers **120** and the Si nano-sheet layers **130**. For example, p-doped Si or Si:C S/D epitaxial layers **140** (e.g., appropriate for n-type FETs) can be grown using a silane (SiH_4) precursor. A phosphorous (P) dopant or carbon can be introduced during growth. In some embodiments, the phosphorous concentration can range from about 1×10^{21} atoms/cm^3 to about 8×10^{21} atoms/cm^3. The aforementioned doping concentration range is not limiting and other doping concentration ranges are within the spirit and the scope of this disclosure.

(46) Accordingly, a B-doped SiGe S/D epitaxial layers **140** (e.g., appropriate for p-type FETs) can include two or more epitaxial layers grown in succession and featuring different Ge atomic percentages and B concentrations. In some embodiments, a first layer can have a Ge atomic % that

ranges from 0 to about 40%, and a B dopant concentration that ranges from about 5×10.sup.19 atoms/cm.sup.3 to about 1×10.sup.21 atoms/cm.sup.3. A second epitaxial layer can have a Ge atomic % that ranges from about 20% to about 80%, and a B dopant concentration that ranges from about 3×10.sup.20 atoms/cm.sup.3 to about 5×10.sup.21 atoms/cm.sup.3. Further, a third epitaxial layer can be a capping layer that has similar Ge atomic % and B dopant concentrations as the first layer (e.g., 0 to about 40% for Ge, and about 5×10.sup.19 atoms/cm.sup.3 to about 1×10.sup.21 atoms/cm.sup.3 for B dopant). The aforementioned doping concentrations are not limiting and other doping concentrations are within the spirit and the scope of this disclosure.
(47) In some embodiments, after the formation of S/D epitaxial layers **140**, ILD **180** is deposited and subsequently planarized (e.g., with a CMP process) so that ILD **180** is substantially co-planar with polysilicon gate electrode **805**. During the aforementioned planarization process, gate capping layer **815**, which functions as a planarization etch stop layer, is removed from the top surface of polysilicon gate electrode **805**. Therefore, after operation **340**, the top surface of polysilicon gate electrode **805** is exposed as shown in FIG. **13**. In some embodiments, ILD **190** includes SiO.sub.2, SiOC, SiON, SiOCN, or SiCN deposited by CVD, physical vapor deposition (PVD), a thermal process, or any other appropriate deposition method.
(48) In referring to FIG. **3**B, method **300** continues with operation **345** and the process of removing polysilicon gate electrode **805** from sacrificial gate structure **800**. Operation **340**, like the previous operations of method **300**, is common between HV-NS FET **100** and LV-GAA FET **200** shown respectively in FIGS. **1** and **2**. In some embodiments, operation **345** includes a wet etching process during which polysilicon gate electrode **805** is selectively removed. In some embodiments, the wet chemistry used in operation **345** has a high selectivity towards polysilicon gate electrode **805** compared to the surrounding layer such as gate dielectric **160***a*, gate spacers **170**, and ILD **180**. By way of example and not limitation, the selectivity of the wet etching chemistry between polysilicon gate electrode **805** and the surrounding materials (e.g., gate dielectric **160***a*, gate spacers **170**, etc.) is greater than about 1,000:1 (e.g., 10,000:1). By way of example and not limitation, the wet etching chemistry can include ammonium hydroxide (NH.sub.4OH). Since gate dielectric **160***a* is interposed between vertical nano-sheet layer structure **700** and polysilicon gate electrode **805**, gate dielectric **160***a* protects the SiGe/Si nano-sheet layers of vertical nano-sheet layer structure **700** from being etched during the etching process.
(49) At this fabrication stage of method **300** (e.g., after operation **340**), partially fabricated HV-NS FETs and LV-GAA FETs share substantially similar features. For example, FIG. **14** shows a partially fabricated structure that can be either a HV-NS FET or a LV-GAA FET. Subsequent operations of method **300** can differentiate HV-NS FETs from LV-GAA FETs because these operations introduce features specific for these FETs.
(50) In referring to FIG. **3**B, method **300** continues with operation **350** and the process of selectively masking partially fabricated HV-NS FETs. For example, a hard mask layer can be disposed on both partially fabricated HV-NS FETs and LV-GAA FETs. The hard mask layer is subsequently patterned to form selectively a masking layer on the partially fabricated HV-NS FETs but not on the partially fabricated LV-GAA FETs. For example, FIG. **15** shows a partially fabricated HV-NS FET **100** covered by a masking layer **1500** and a partially fabricated LV-GAA FET **200** not covered by masking layer **1500** according to operation **350**. By way of example and not limitation, masking layer **1500** can be a nitride layer, a metal oxide layer, or a photoresist layer.
(51) In referring to FIG. **3**B, method **300** continues with operation **355** and the process of removing SiGe nano-sheet layers **120** from vertical nano-sheet layer structure **700** of LV-GAA FETs **200**. In some embodiments, operation **355** is reserved for LV-GAA FETs **200** since HV-NS FETs **100** are “protected” by masking layer **1500**. In some embodiments, prior to removing SiGe nano-sheet layers **120**, gate dielectric **160***a* is removed with an etching process (e.g., a wet etching process) from exposed LV-GAA FETs **200**. In some embodiments, the etching process does not remove portions of gate dielectric **160***a* covered by gate spacers **170** as shown in FIG. **16**.

(52) Removal of SiGe nano-sheet layers **120** is achieved, for example, with an etching chemistry similar to the one used in operation **325** for the lateral etching of SiGe nano-sheet layers **120**. For example, in operation **350**, SiGe nano-sheet layers **120** can be exposed to a halogen-based chemistry (e.g., fluorine-based and/or chlorine-based gases) until SiGe nano-sheet layers **120** are completely removed from vertical nano-sheet structure **700**. After the removal of SiGe nano-sheet layers **120**, Si nano-sheet layers **130** become suspended between S/D epitaxial layers **140** as shown in FIG. **16**.
(53) In some embodiments, surfaces of Si nano-sheet layers **130** may appear recessed (e.g., thinned) after operation **355** as shown in FIG. **17**, which is a magnified view of section **1600** shown in FIG. **16**. For example, middle portion thickness Mr can be equal to or shorter than edge portion thickness E_T of Si nano-sheet layers **130** (e.g., $E_T \geq M_T$). In some embodiments, the height difference between the middle and edge portions of Si nano-sheet layers **130** after operation **350** can be between about 2 nm and about 4 nm—for example, ($E_T - M_T$) is between about 2 nm and about 4 nm. In some embodiments, the aforementioned "thinning" of Si nano-sheet layers **130** during operation **355** is attributed to the selectivity of the etching gasses used in operation **355**. For example, the etching selectivity of the etching chemistry towards Si nano-sheet layers **130** may not be zero.
(54) In some embodiments, after operation **355**, masking layer **1500** shown in FIG. **15** is removed over HV-NS FETs **100** as shown in FIG. **18**. Therefore, after operation **355**, HV-NS FETs **100** are un-masked so that subsequent processing can be equally applied to both HV-NS FETs **100** and LV-GAA FETs **200**.
(55) In referring to FIG. **3**B, method **300** continues with operation **360** and the process of forming gate structure **160** in HV-NS FETs **100** and LV-GAA FETs **200**. Since SiGe nano-sheet layers **120** have been removed from LV-GAA FETs **200**, gate structure **160** will surround each Si nano-sheet layer **120** as shown in FIG. **19**. At the same time, gate structure **160** will cover sidewall and top surfaces of vertical nano-sheet layer structure **700**.
(56) In some embodiments, IL **160***d* is formed first, followed by high-k gate dielectric **160***b*, and gate electrode **160***c*. In some embodiments, IL **160***d* is not distinguishable from gate dielectric **160***a* in HV-NS FETs **100** as shown in FIGS. **1** and **19**. This is because IL **160***d* and gate dielectric **160***a* can be made from the same material. On the other hand, IL **160***d* is distinguishable in LV-GAA FETs **200** as shown in FIGS. **2** and **19**.
(57) In some embodiments, IL **160***d* includes silicon oxide or a silicon oxy-nitride and high-k gate dielectric **160***b* includes doped or un-doped hafnium oxide (HfO_2), a hafnium silicate-based material, or another suitable dielectric material with a k-value greater than about 3.9. Further, gate electrode **160***c* can include a capping layer disposed on high-k gate dielectric **160***b*, one or more barrier layers, a work-function metal (WFM) stack, and a metal fill layer. The aforementioned layers of gate electrode **160***c* are not shown in FIGS. **1**, **2**, and **19** for simplicity. The number and type of layers in the WFM stack in gate electrode **160***c* modulates the threshold voltage of HV-NS FETs **100** and, most importantly, the threshold voltage of LV-GAA FETs **200**. In some embodiments, the WFM stack includes tantalum nitride (TaN) layers, titanium nitride (TiN) layers, (Ti/Al) bi-layers, titanium-aluminum (Ti—Al) alloy layers, tantalum-aluminum (Ta—Al) alloy layers, or combinations thereof. In some embodiments, metal fill layer can include a TiN barrier layer and a tungsten (W) metal stack. The aforementioned list of materials for IL **160***d*, high-k gate dielectric **160***b*, and gate electrode **160***c* is not exhaustive. Therefore, additional materials can be used. These additional materials are within the spirit and the scope of this disclosure.
(58) In some embodiments, after the formation of gate structure **160**, a planarization process (e.g., a CMP process) removes deposited gate stack material on ILD **180** and substantially planarizes HV-NS FETs **100** and LV-GAA FETs **200** so that gate structure **160** and ILD **180** are substantially coplanar as shown in FIGS. **1**, **2**, and **19**. Subsequently, S/D contacts and silicide layers can be formed on S/D epitaxial layers **140**. These S/D contacts and silicide layers are not shown in FIGS.

1, **2**, and **19** for simplicity.
(59) In some embodiments, the HV-NS FETs described above can be used in a cascode arrangement (e.g., a two-stage amplifier that includes a common-emitter stage feeding into a common-base stage) to form high-voltage drivers for an OTP circuitry. According to some embodiments, FIG. **20**A is a partial view of an OTP circuitry, where a high-voltage driver formed by 4 HV-NS FETs in a cascode arrangement (e.g., a 2× cascode) is connected to a bit-cell array via a programming word-line (WLP). In some embodiments, the HV-NS FETs arranged as a high-voltage driver, as compared to HV planar FETs (e.g., without nano-sheet layers), can reduce the area of the high-voltage driver between about 25% and about 50%. This is because the footprint of HV-NS FETs is smaller than that of a HV planar FET. For example, a 4× cascode (e.g., a cascode arrangement of 8 FETs) of HV planar FETs can be reduced to a 3× cascode (e.g., a cascode arrangement with 6 FETs) or a 2× cascode (e.g., a cascode arrangement with 4 FETs, like the one shown in FIG. **18**A) with HV-NS FETs.
(60) According to some embodiments, FIG. **20**B is another exemplary OTP circuit where the bit-cell array, as opposed to the bit-cell array of FIG. **20**A, includes HV-NS FETs. In some embodiments, the benefit of the bit-cell array configuration shown in FIG. **20**B is an increase in the read time (e.g., increased read time by 3×) compared to the configuration shown in FIG. **20**A.
(61) The embodiments described herein are directed to methods for the formation of HV-NS FETs and LV-GAA FETs on a common substrate. In some embodiments, HV-NS FETs and LV-GAA FETs are fabricated concurrently. In some embodiments, the HV-NS FETs operate at about 5 Volts and the LV-GAA FETs operate at about 1.8 volts or less. The HV-NS FETs as described herein include a gate dielectric stack with a silicon oxide layer having a thickness between about 1.5 nm and 3.5 nm and a high-k dielectric layer disposed thereon. In some embodiments, HV-NS FETs and LV-GAA FETs share the same gate electrode. According to some embodiments, the HV-NS FETs include a fin structure with alternating Si and SiGe nano-sheet layers. In some embodiments, and during operation, current conduction in the HV-NS FETs occurs within the silicon nano-sheet layers. Further, both HV-NS FETs and LV-GAA FETs share similar S/D epitaxial structures.
(62) In some embodiments, a structure includes a substrate and a fin structure on the substrate that includes alternating first and second nano-sheet layers. The structure further includes an epitaxial structure abutting end-portions the fin structure and a spacer structure interposed between the first nano-sheet layers of the fin structure and the epitaxial structure. The structure also includes a gate structure on a top surface and sidewall surfaces of the fin structure, where the gate structure includes a first dielectric and a second dielectric, thinner than the first dielectric, disposed on the first dielectric.
(63) In some embodiments, a circuitry includes a substrate with a first device on the substrate, includes a fin structure with alternating first and second nano-sheet layers and first S/D epitaxial structures abutting end-portions of the fin structure, where each first S/D epitaxial structure is electrically isolated from the first nano-sheet layers. The first device also includes a first gate structure on the fin structure that inlcudes a first dielectric and a second dielectric. The circuitry further includes a second device on the substrate having the second nano-sheet layers between second S/D epitaxial structures. The second device also includes a second gate structure surrounding the second nano-sheet layers between the second S/D epitaxial structures, where the second gate structure includes the second dielectric.
(64) In some embodiments, a method includes forming a fin structure on a substrate, where the fin structure includes first and second nano-sheet layers. The method also includes forming a gate structure on the fin structure with the gate structure having a first dielectric and a first gate electrode, and removing portions of the fin structure not covered by the gate structure. The method further includes partially etching exposed surfaces of the first nano-sheet layers to form recessed portions of the first nano-sheet layers in the fin structure, depositing a capping layer on the fin structure to fill the recessed portions, and etching the capping layer from the fin structure to form a

spacer structure on the recessed portions. In addition, the method includes replacing the first gate electrode with a second dielectric and a second gate electrode, and forming an epitaxial structure abutting the fin structure so that the epitaxial structure is in physical contact with the first nano-sheet layers and isolated from the second nano-sheet layers by the spacer structure.
(65) It is to be appreciated that the Detailed Description section, and not the Abstract of the Disclosure section, is intended to be used to interpret the claims. The Abstract of the Disclosure section may set forth one or more but not all possible embodiments of the present disclosure as contemplated by the inventor(s), and thus, are not intended to limit the subjoined claims in any way.
(66) The foregoing disclosure outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art will appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art will also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A method, comprising: forming a stack of first and second nanostructured layers on a substrate; forming a gate structure comprising a first dielectric layer and a first gate electrode on the stack of first and second nanostructured layers; forming a gate spacer on the first dielectric layer; forming a gate capping layer directly on top surfaces of the gate structure and the gate spacer; removing portions of the first and second nanostructured layers uncovered by the gate structure, wherein sidewalls of the first dielectric layer, the gate capping layer, and the gate spacer are coplanar with sidewalls of the first nanostructured layers along a vertical plane after removing portions of the first and second nanostructured layers; etching exposed surfaces of the second nanostructured layers to form recessed regions in the second nanostructured layers; depositing a capping layer to fill the recessed regions; etching the capping layer to form spacer structures on the recessed regions; forming an epitaxial structure abutting the spacer structures and the first nanostructured layers; and replacing the first gate electrode with a second dielectric layer and a second gate electrode on the first dielectric layer.
2. The method of claim 1, wherein etching the exposed surfaces of the second nanostructured layers comprises reducing lengths of the second nanostructured layers.
3. The method of claim 1, wherein replacing the first gate electrode comprises etching the first gate electrode and exposing a top surface of the first dielectric layer.
4. The method of claim 1, wherein replacing the first gate electrode with the second dielectric layer comprises depositing the second dielectric layer with a thickness less than a thickness of the first dielectric layer.
5. The method of claim 1, wherein forming the first nanostructured layers comprises epitaxially growing semiconductor layers with a doping concentration of about 10.sup.13 atoms/cm.sup.3.
6. The method of claim 1, wherein forming the second nanostructured layers comprises epitaxially growing semiconductor layers with a germanium concentration between about 20% and about 30%.
7. The method of claim 1, wherein etching the exposed surfaces of the second nanostructured layers comprises performing a lateral etch on sidewalls of the second nanostructured layers.
8. The method of claim 1, wherein depositing the capping layer comprises depositing a silicon nitride based dielectric layer.
9. A method, comprising: forming a first stack of first and second nanostructured layers on a

substrate, wherein first horizontal dimensions of the first and second nanostructured layers along a first cross-sectional plane are different from each other, and wherein second horizontal dimensions of the first and second nanostructured layers along a second cross-sectional plane are equal to each other; forming a second stack of third and fourth nanostructured layers on the substrate; forming a first gate structure comprising a first oxide layer and a first polysilicon layer on the first stack of first and second nanostructured layers; forming a second gate structure comprising a second oxide layer and a second polysilicon layer on the second stack of third and fourth nanostructured layers; forming first and second gate spacers on the first and second oxide layers, respectively; forming a gate capping layer directly on top surfaces of the first polysilicon layer and the first gate spacer, wherein sidewalls of the first oxide layer, the gate capping layer, and the first gate spacer are aligned with sidewalls of the first nanostructured layer along a vertical direction; replacing the first polysilicon layer with a first high-k dielectric layer directly on the first oxide layer; and replacing the second polysilicon layer, the second oxide layer, and the fourth nanostructured layers with a second high-k dielectric layer to surround the third nanostructured layers.
10. The method of claim 9, wherein replacing the second polysilicon layer, the second oxide layer, and the fourth nanostructured layers comprises removing the fourth nanostructured layers after removing the second polysilicon layer and the second oxide layer.
11. The method of claim 9, wherein replacing the second polysilicon layer, the second oxide layer, and the fourth nanostructured layers comprises etching the second polysilicon layer, the second oxide layer, and the fourth nanostructured layers to expose top and bottom surfaces of the third nanostructured layers.
12. The method of claim 9, further comprising oxidizing top and bottom surfaces of the third nanostructured layers.
13. The method of claim 9, wherein replacing the first polysilicon layer comprises etching the first polysilicon layer to expose a top surface of the first oxide layer, and further comprising forming a masking layer on the top surface of the first oxide layer.
14. The method of claim 9, replacing the first polysilicon layer with the first high-k dielectric layer comprises depositing the first high-k dielectric with a thickness less than a thickness of the first oxide layer.
15. The method of claim 9, wherein replacing the second polysilicon layer, the second oxide layer, and the fourth nanostructured layers comprises thinning middle portions of the third nanostructured layers.
16. The method of claim 9, further comprising depositing first and second gate electrodes on the first and second high-k dielectric layers, respectively.
17. A method, comprising: forming first and second nanostructured semiconductor layers on a substrate, wherein first horizontal dimensions of the first and second nanostructured semiconductor layers along a first cross-sectional plane are different from each other, and wherein second horizontal dimensions of the first and second nanostructured semiconductor layers along a second cross-sectional plane are equal to each other; forming third and fourth nanostructured semiconductor layers on the substrate; forming first and second oxide layers on the first and third nanostructured semiconductor layers, respectively; forming first and second polysilicon layers on the first and second oxide layers, respectively; forming first and second gate spacers on the first and second oxide layers, respectively; forming a gate capping layer directly on top surfaces of the first polysilicon layer and the first gate spacer, wherein sidewalls of the first oxide layer, the gate capping layer, and the first gate spacer are aligned with sidewalls of the first nanostructured semiconductor layers along a vertical direction; removing the first polysilicon layer to expose a top surface the first oxide layer; removing the second polysilicon layer and the second oxide layer to expose a top surface of the third nanostructured semiconductor layer; removing the fourth nanostructured semiconductor layer to expose a bottom surface of the third nanostructured semiconductor layer; and depositing first and second gate electrodes on the first oxide layer and the

third nanostructured semiconductor layer, respectively.
18. The method of claim 17, further comprising forming a masking layer on the top surface the first oxide layer after removing the first polysilicon layer.
19. The method of claim 17, further comprising depositing first and second high-k gate dielectric layers on the first oxide layer and the third nanostructured semiconductor layer, respectively.
20. The method of claim 17, further comprising forming spacer structures on sidewalls of the second nanostructured semiconductor layers.